

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
1 | #include <stdio.h>
```

```
1 // C++ program to find a pair with difference k
2
3 int main() {
4     int n, k;
5     scanf("%d", &n);
6
7     int a[n];
8
9     for (int i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     scanf("%d", &k);
13
14     int i = 0, j = 1;
15
16     while (i < n && j < n) {
17         if (i == j) {
18             j++;
19         }
20         else if (a[j] - a[i] == k) {
21             printf("1");
22             return 0;
23         }
24         else if (a[j] - a[i] < k) {
25             j++;
26         }
27         else {
28             i++;
29         }
30     }
31
32     printf("0");
33
34     return 0;
35 }
```